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Experiment 1: Case Study

Problem Statement: Choose a real-world case study such as an KL ERP System, create user stories, select an appropriate software development process model, identify a suitable programming language, and propose an implementation and evaluation strategy.

Real-World case study an online shopping platform

1. Functional Requirements

Faculty

- Login securely
- View assigned courses
- Record student attendance
- Update attendance records
- Upload course materials
- Create assignments
- Enter internal and examination marks
- View student performance
- Send announcements to students
- View teaching timetable
- Generate class performance reports

Administrator

- Login securely
- Manage student records
- Manage faculty records
- Manage course information
- Assign faculty to courses
- Create academic timetables
- Manage examination schedules
- Generate institutional reports
- Manage user accounts and permissions
- Monitor fee payment records
- Backup and restore system data

2. User Stories

User ID	User Story	Priority
US-01	As a faculty member, I want to log in securely so that I can access the ERP system safely.	High 8

User ID	User Story	Priority
US-02	As a faculty member, I want to view my assigned courses so that I can manage my teaching schedule.	High 6
US-03	As a faculty member, I want to record student attendance so that attendance records remain accurate.	High
US-04	As a faculty member, I want to upload lecture notes and course materials so that students can access learning resources..	High
US-05	As a faculty member, I want to enter internal and examination marks so that students can view their academic performance.	Medium2
US-06	As a faculty member, I want to send announcements to students so that they receive important academic updates.	High
US-07	As a faculty member, I want to create assignments so that students can complete coursework online.	High
US-08	As an administrator, I want to manage student records so that student information remains accurate and up to date.	Medium
US-09	As an administrator, I want to assign faculty members to courses so that teaching responsibilities are organized properly.	High
US-10	As an administrator, I want to generate academic and administrative reports so that management can monitor institutional performance.	High
US-11	As an administrator, I want to manage user accounts and permissions so that only authorized users can access the system.	Medium
US-12	As an administrator, I want to manage examination schedules so that exams are conducted without conflicts.	Medium

3. Selected Software Development Process Model

Agile Scrum

Reason for Selection

- Requirements change frequently in educational institutions.
- Continuous feedback from faculty and administrators.
- Incremental delivery of ERP modules.
- Faster development and testing.
- Easy maintenance and bug fixing.
- Suitable for large-scale academic management systems.

Sprint Plan

Sprint	Features
Sprint 1	User Authentication & Login
Sprint 2	Faculty Dashboard & Course Management
Sprint 3	Attendance Management
Sprint 4	Examination & Marks Management
Sprint 5	Student & Faculty Record Management
Sprint 6	Reports & Notifications
Sprint 7	Testing & Deployment

4. Programming Language Selection

Backend

- Java (Spring Boot)

Reasons

- Secure
- Scalable
- Enterprise-level applications
- REST API support
- Excellent database connectivity

Frontend

- HTML5
- CSS3
- JavaScript
- React.js

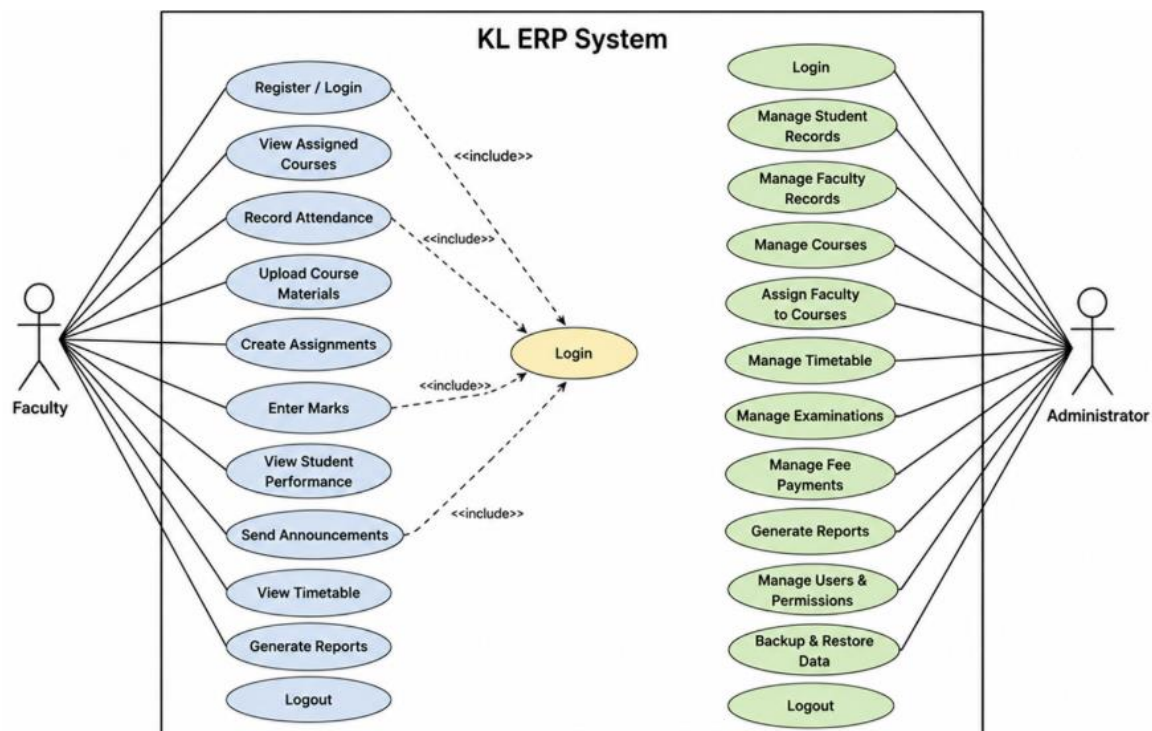
Database

- MySQL

Other Technologies

- REST API
- JWT Authentication
- Stripe/Razorpay Payment Gateway
- Git
- Docker
- AWS Cloud

5. System Architecture



6. Implementation Strategy

Phase 1 – Requirement Analysis

- Gather requirements from faculty and administrators.
- Prepare the Software Requirement Specification (SRS).
- Develop use case diagrams.
- Identify system modules.

Phase 2 – System Design

- Design database schema.
- Design user interface wireframes
- Create UML diagrams.
- Develop overall system architecture.

Phase 3 – Development

- Implement secure login and authentication.
- Develop Faculty Dashboard.
- Develop Attendance Management Module.
- Develop Examination & Marks Module.
- Develop Student & Faculty Management Module
- Develop Reports and Notification Module.

Phase 4 – Testing

Unit Testing

- Login
- Attendance
- Marks Entry
- Student Record Management

Integration Testing

- Attendance + Reports
- Faculty + Examination Module
- User Authentication + Database

System Testing

- Complete ERP functionality

User Acceptance Testing (UAT)

- Verify all academic and administrative requirements.

Phase 5 – Deployment

- Docker Container
- AWS EC2
- MySQL Database
- Nginx Web Server

7. Evaluation Strategy

Evaluation Parameter Method

Functional Accuracy	Requirement verification
Performance	Load testing using JMeter
Security	Authentication and penetration testing
Reliability	Continuous monitoring
Usability	User feedback survey
Scalability	Concurrent user testing
Maintainability	Code review and documentation

10. Expected Outcomes

- Secure online shopping platform.
- Fast product search.
- Reliable order management.

- Accurate inventory updates.
- Secure online payments.
- Improved customer satisfaction.
- Scalable architecture for future enhancements.